Facundo Hernandez

TP 1

Binario a decimal Ejercicio 1)

a) 001100 $= 0.2^5 + 0.2^4 + 1.2^3 + 1.2^2 + 0.2^1 + 0.2^0$

5 4 3 2 1 0

$0 + 0 + 8 + 4 + 0 + 0$

12

b) 000011 $= 0.2^5 + 0.2^4 + 0.2^3 + 0.2^2 + 1.2^1 + 1.2^0$

5 4 3 2 1 0

$0 + 0 + 0 + 0 + 2 + 1$

3

c) 011100 $= 0.2^5 + 1.2^4 + 1.2^3 + 1.2^2 + 0.2^1 + 0.2^0$

5 4 3 2 1 0

$0 + 16 + 8 + 4 + 0 + 0$

28

d) 111100 $= 1.2^5 + 1.2^4 + 1.2^3 + 1.2^2 + 0.2^1 + 0.2^0$

5 4 3 2 1 0

$32 + 16 + 8 + 4 + 0 + 0$

60

e) 101010 $= 1.2^5 + 0.2^4 + 1.2^3 + 0.2^2 + 1.2^1 + 0.2^0$

5 4 3 2 1 0

$32 + 0 + 8 + 0 + 2 + 0$

42

Ejercicio 2) a) 64 |2

0 32 |2

0 16 |2

0 8 |2

0 4 |2

0 2 |2

0 1 |2

1 0

Valor binario: 1000000

b) 100 | 2

0 50 | 2

0 25 | 2

1 12 | 2

0 6 | 2

0 3 | 2

1 1 | 2

1 0

Binario: 1100100

c) 111 | 2

1 55 | 2

1 27 | 2

1 13 | 2

1 6 | 2

0 3 | 2

1 1 | 2

1 0

Binario: 1101111

d) 145 | 2

1 72 | 2

0 36 | 2

0 18 | 2

0 9 | 2

1 4 | 2

0 2 | 2

0 1 | 2

1 0

Binario: 10010001

e) 225 | 2

1 112 | 2

0 56 | 2

0 28 | 2

0 14 | 2

0 7 | 2

1 3 | 2

1 1 | 2

1 0

Binario: 11100001

Ejercicio 3) a) 12 Paso a binario

1 2

001 010

$\underbrace{00}_{0}\,\underbrace{001010}_{A}$

$\boxed{0A}$

b) 5655

5 6 5 5

101 110 101 101

$\underbrace{1011}_{B}\underbrace{1010}_{A}\underbrace{1101}_{D} = \boxed{BAD}$

c) 2550276

2 5 5 0 2 7 6

$\underbrace{000\,0}_{0}\underbrace{1010}_{A}\underbrace{1101}_{D}\underbrace{0000}_{0}\underbrace{1011}_{B}\underbrace{1110}_{E}$ $\boxed{ADOBE}$

d) 76545336

7 6 5 4 5 3 3 6

$\underbrace{1111}_{F}\underbrace{1010}_{A}\underbrace{1100}_{C}\underbrace{1010}_{A}\underbrace{1101}_{D}\underbrace{1110}_{E}$

e) 3726755   3 7 2 6 7 5 5

$\underbrace{000}\underbrace{011}\underbrace{111}\underbrace{010}\underbrace{110}\underbrace{111}\underbrace{101}\underbrace{101}$

0 F A D E D

Ejercicio 4) a) C_0 $12.16^0 = 12$ b) $9_1 F_0$ $9.16^1 + 15.16^0$

144 + 15

159

c) $D_2 5_1 2_0$ $13.16^2 + 5.16^1 + 2.16^0$ d) $6_2 7_1 E_0$ $6.16^2 + 7.16^1 + 14.16^0$

3,328 + 80 + 2   1536 + 112 + 14

3410   1662

e) $A_3 B_2 C_1 D_0$ $10.16^3 + 11.16^2 + 12.16^1 + 13.16^0$

40,960 + 2,816 + 192 + 13

43,981

Ejercicio 5) 16 | 16   b) 80 | 16

0 1 | 16   0 5 | 16 R: 50

1 0 R: 10   5 0

c) 2560 | 16

0 160 | 16

0 10 | 16 R: A00

10 0

d) 3000 | 16

8 187 | 16

11 11 | 16 R: BB8

11 0

e) 62500 | 16

4 3906 | 16

2 244 | 16 R: F424

4 15 | 16

15 0

Ejercicio 6: a) E → 1110 b) 1C → 1 1100 11100

c) A64 → 101001100100 d) 1FC → 1 1111 1100 111111100

e) 2394 → 10 0011 1001 0100 R: 10001110010100

Ejercicio 7) $34,75_{(10)}$

34 | 2
0 17 | 2
1 8 | 2
0 4 | 2
0 2 | 2
0 1 | 2
1 0

$0,75 \cdot 2 = 1,5$
$0,5 \cdot 2 = 1,0$

$\boxed{Rta: 100010,11}$

b) 25,25

25 | 2
1 12 | 2
0 6 | 2
0 3 | 2
1 1 | 2
1 0

$0,25 \times 2 = 0,5$
$0,5 \times 2 = 1,0$

$\boxed{Rta: 11001,01}$

c) 27,1875

27 | 2
1 13 | 2
1 6 | 2
0 3 | 2
1 1 | 2
1 0

$0,1875 \cdot 2 = 0,375$
$0,375 \cdot 2 = 0,75$
$0,75 \cdot 2 = 1,5$
$0,5 \cdot 2 = 1,0$

$\boxed{Rta: 11011,0011}$

d) 45,625

$45 \,|\, 2$
1 $22 \,|\, 2$
0 $11 \,|\, 2$
1 $5 \,|\, 2$
1 $2 \,|\, 2$
0 $1 \,|\, 2$
1 0

$0,625 * 2 = 1,25$
$0,25 * 2 = 0,5$
$0,5 * 2 = 1.0$

$\boxed{\text{Rta: } 101101,101}$

e) 129,2

$129 \,|\, 2$
1 $64 \,|\, 2$
0 $32 \,|\, 2$
0 $16 \,|\, 2$
0 $8 \,|\, 2$
0 $4 \,|\, 2$
0 $2 \,|\, 2$
0 $1 \,|\, 2$
1 0

$0,2 * 2 = 0,4$
$0,4 * 2 = 0,8$
$0,8 * 2 = 1,6$
$0,6 * 2 = 1,2$
$0,2$ = SE REPITE

$\boxed{\text{Rta: } 10000001,0011}$

Ejercicio 8: a) 204.125

$204 \,|\, 16$
12 $12 \,|\, 16$
12 0

$0,125 * 16 = 2.0$

$\boxed{\text{Rta: CC,2}}$

b) $255 \,|\, 16$
15 $15 \,|\, 16$
15 0

$0,875 * 16 = 14.0$

$\boxed{\text{Rta: FF.E}}$

c) 631,25

$631 \,|\, 16$
7 $39 \,|\, 16$
7 $2 \,|\, 16$
2 0

$0,25 * 16 = 4$

$\boxed{\text{Rta: } 277,4}$

d) 10000,00390625   0,00390625*16 = 0,0625
0,0625*16 = 1

10000 |16
0 625 |16
1 39 |16
7 2 |16
2 0

$\boxed{\text{Rta}: 2710,01}$